

6(a)(i)	greater information carrying capacity	B1
6(a)(ii)	power/energy is radiated	B1
	signal picked up by adjacent fibre/wire	B1
6(b)	ratio / dB = $10 \lg(P_2 / P_1)$	C1
	$13 = 10 \lg [P / (1.0 \times 10^{-3})]$ and so $P = 20 \text{ mW}$	A1
6(c)	$45 \times 0.18 = 10 \lg (20 / P)$	C1
	$P = 3.1 \text{ mW}$	A1